

Smart Plant Monitoring Bot

Submitted as a Final Lab Project Report for the Course Robotics Lab
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by

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Abstract

Plant protection and monitoring are important challenges in agriculture, home gardening, and greenhouse environments. Traditional monitoring methods often require continuous human supervision, which is time-consuming and inefficient. This project presents a Smart Plant Monitoring Bot that integrates robotics, embedded systems, and automated security monitoring technologies to protect plants from potential threats such as humans, animals, birds, and other moving objects.

The system uses Arduino UNO as the main controller, IR line tracking sensors for autonomous navigation, an ultrasonic sensor for obstacle detection, a PIR motion sensor for threat monitoring, a servo motor for environmental scanning, a 16×2 I2C LCD for status display, a GSM module for SMS notifications, a buzzer for alarm generation, and DC motors for robot movement. The robot continuously patrols a predefined path using line-following technology. When an obstacle is detected, the robot stops, scans the surrounding area using the ultrasonic sensor mounted on the servo motor, selects a clear path, and resumes patrolling. If motion is detected near the protected plants, the robot immediately stops, activates an alarm, displays warning information and distance measurements on the LCD, and sends an SMS alert to the owner through the GSM module.

The prototype was tested under various operating conditions, including line-following navigation, obstacle avoidance, and threat detection scenarios. The results demonstrated successful autonomous movement, reliable obstacle avoidance, accurate motion detection, real-time alert generation, LCD status updates, and GSM-based notification delivery. The proposed system reduces the need for continuous human supervision, improves plant protection efficiency, and demonstrates a practical application of robotics and embedded systems in smart agriculture and automated plant monitoring.

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Chapter 1

Introduction

In present-day agriculture and domestication, when beasts and insects are trying to raid, it is important to safeguard plants. and environmental disturbances is an important challenge. Farmers and gardeners Players can often encounter difficulties when birds, cats, dogs, insects or even mistake human interference!activities damage plants. Traditional monitoring methods require continuous human The time-consuming and inefficient supervision.

There is a feasible solution for auto- embedded system technologies and robotics. mated monitoring and protection of plants. There are sensors, micro-controllers and commu- nication modules can co-operatively detect threats, keep an eye on the environment, Send alerts automatically.

This project presents a **Smart Plant Monitoring Bot** designed using Arduino. Various sensors for detection: UNO, IR line tracking sensors, ultrasonic sensors, PIR motion sensor, GSM communication, nication, LCD display, servo motor, buzzer and DC motors. The robot continuously Move patrols (using line following technology) along a defined path. It can detect and Limit motion to avoid obstacles and automatically track plant surroundings for potential hazards threats.

Once motion is detected near the protected plants, motion sensing immediately turns off. engages boy, triggers an alarm, lights up alerting data on the LCD, records Records distance of detected object, and sends SMS notification to own. via the GSM module. Once the threat has passed, the robot returns to normal. patrolling

operation.

The proposed solution is a combination of robotics, embedded systems, automation and security. Monitoring to a low cost prototype suitable for agricultural and smart gardening. applications.

1.1 Background and Motivation

Plant protection is a significant concern in agriculture, nurseries, greenhouses, and home gardening environments. Animals such as dogs, cats, birds, and insects can damage plants and reduce productivity. Continuous monitoring by humans is often impractical, especially in large areas.

Recent developments in robotics and embedded systems have enabled the creation of autonomous monitoring systems capable of performing surveillance tasks with minimal human intervention. Mobile robots equipped with sensors can patrol designated areas, detect threats, avoid obstacles, and communicate alerts in real time.

The motivation behind this project is to develop an affordable and autonomous plant monitoring system that can patrol continuously, detect potential threats, and notify the owner whenever unusual activity is detected around the plants.

1.2 Problem Statement

Traditional plant protection methods depend heavily on human observation and manual intervention. This creates several limitations:

- Continuous monitoring is difficult and time-consuming
- Animal attacks may occur when nobody is present.
- Obstacle-filled environments can interrupt surveillance activities.
- Immediate notification systems are often unavailable.

- Manual protection methods are inefficient for large gardens or farms.

Therefore, there is a need for an intelligent robotic system capable of autonomous navigation, obstacle avoidance, threat detection, and real-time alert generation.

1.3 Objectives of the Project

The main objective of this project is to design and implement a Smart Plant Monitoring Bot capable of autonomous plant surveillance and protection.

The specific objectives are:

1. To design a mobile robot using Arduino UNO.
2. To implement line-following navigation using IR sensors.
3. To detect and avoid obstacles using ultrasonic sensors and a servo motor.
4. To monitor plant surroundings using a PIR motion sensor.
5. To generate audio alerts using a buzzer during threat detection.
6. To display system status and warning messages on a 16×2 I2C LCD.
7. To send SMS notifications using a GSM module.
8. To automatically resume patrol operation after threat removal.
9. demonstrate the practical application of robotics in plant protection.

1.4 Research Questions

The project is guided by the following research questions:

1. How effectively can a mobile robot monitor plants autonomously?
2. How accurately can IR sensors perform line-following navigation?

3. How efficiently can ultrasonic sensors avoid obstacles during patrol?
4. Can PIR sensors reliably detect nearby threats?
5. How effective is GSM communication for real-time alert generation?
6. Can autonomous monitoring reduce the need for continuous human supervision?

1.5 Scope of the Work

This project focuses on developing a prototype-level Smart Plant Monitoring Bot using embedded systems and robotics technologies.

The scope includes:

- Autonomous line-following patrol.
- Obstacle detection and avoidance.
- Motion detection around plants.
- Audio alarm generation.
- LCD-based status display.
- GSM-based SMS notification.

The project does not include:

- Camera-based object recognition.
- AI-based classification of animals.
- Internet-based cloud monitoring.
- GPS navigation.
- Machine learning algorithms.
- Large-scale agricultural deployment.

These features may be added in future versions

1.6 Expected Outcomes

The expected outcome is a functional robotic prototype capable of:

- Following a predefined path automatically.
- Detecting and avoiding obstacles.
- Monitoring plant surroundings continuously.
- Detecting motion near protected plants.
- Generating audible alerts.
- Displaying warning information on an LCD.
- Sending SMS alerts to the owner.
- Resuming patrol operation automatically after threat removal.

The system is expected to improve plant security and reduce manual monitoring requirements.

1.7 Impacts of the Project

The proposed system contributes to smart agriculture and automated monitoring by reducing manual effort and improving plant protection efficiency.

1.7.1 Impact on Societal and Cultural Issues

The project promotes the use of robotics and automation in agriculture and gardening. It encourages technology-based solutions for plant protection and supports awareness of smart farming concepts.

1.7.2 Impact on Safety, and Legal Issue

The system improves safety by providing early warning of unwanted movement around protected plants. Since the system only detects motion and sends alerts, it does not pose significant health risks. GSM communication should be used responsibly to protect user privacy.

1.7.3 Impact on Environment and Sustainability Issues

The system supports sustainable agriculture by helping reduce plant damage and minimizing the need for constant human supervision. It uses low-power electronic components suitable for energy-efficient operation.

1.8 Report Organization

The report is organized as follows:

Chapter 1: Introduction

Presents the background, motivation, problem statement, objectives, scope, expected outcomes, impacts, and report organization.

Chapter 2: Related Works

Reviews existing smart agriculture systems, plant monitoring robots, obstacle avoidance systems, and GSM-based alert systems.

Chapter 3: System Analysis and Design

Describes system requirements, architecture, hardware setup, circuit design, and robot structure.

Chapter 4: Methodology and Implementation

Explains development methodology, tools, algorithms, flowcharts, and implementation details.

Chapter 5: Results and Evaluation

Presents testing procedures, performance analysis, experimental results, and limitations.

Chapter 6: Project Planning and Budget

Includes timeline planning, resource allocation, and budget estimation.

Chapter 7: Standards and Design Constraints

Discusses standards, safety considerations, design constraints, CEP, CEA, and knowledge profile mapping.

Chapter 8: Conclusion and Future Work

Summarizes achievements, limitations, and future development possibilities.

Chapter 2

Related Works

The development of smart agriculture and automated monitoring systems has increased significantly in recent years. Researchers and engineers have proposed various robotic and IoT-based solutions for monitoring crops, protecting plants, detecting intrusions, and automating agricultural tasks. The integration of sensors, microcontrollers, wireless communication modules, and autonomous robots has made agricultural monitoring more efficient and reliable.

This chapter reviews existing works related to plant monitoring systems, agricultural robots, obstacle avoidance robots, motion detection systems, and GSM-based alert systems. The review helps identify the strengths and limitations of existing solutions and highlights the motivation for developing the proposed Smart Plant Monitoring Bot.

2.1 Smart Agriculture Monitoring Systems

TSmart agriculture focuses on the use of technology to improve farming efficiency, crop protection, and resource management. Various monitoring systems have been developed using sensors and wireless communication technologies to observe environmental conditions and detect potential threats.

Many smart agriculture systems use temperature, humidity, soil moisture, and motion sensors to monitor crop conditions. These systems provide farmers with real-time

information and reduce the need for manual inspection. However, most of these systems are fixed installations and cannot patrol large areas autonomously.

The proposed Smart Plant Monitoring Bot differs from conventional monitoring systems because it combines mobility, obstacle avoidance, threat detection, and alert generation in a single robotic platform.

2.2 Plant Protection and Surveillance Robots

Several researchers have developed mobile robots for agricultural monitoring and crop protection. These robots can move through agricultural fields, collect data, and monitor plant conditions.

Plant surveillance robots commonly use cameras, ultrasonic sensors, and navigation systems to move through predefined routes. Some robots focus on pest detection, while others monitor plant growth and environmental conditions.

Although these robots provide useful monitoring capabilities, many of them require expensive hardware, complex image processing systems, or internet connectivity. Such requirements increase the overall cost and complexity of the system.

The proposed Smart Plant Monitoring Bot uses low-cost components while maintaining essential features such as autonomous patrol, obstacle avoidance, motion detection, and SMS-based alerts.

2.3 Line Following Robots

Line-following robots are widely used in industrial automation, warehouse transportation, and robotic navigation applications. These robots use infrared (IR) sensors to detect a predefined path and navigate autonomously.

IR sensors detect the contrast between black and white surfaces. Based on sensor readings, the robot adjusts its movement to remain on the designated path.

Line-following robots provide a simple and reliable navigation mechanism. However, traditional line-following robots are mainly designed for movement and transportation tasks and often lack security monitoring or threat detection capabilities.

The proposed system extends conventional line-following technology by integrating plant monitoring and protection functionalities.

2.4 Obstacle Avoidance Systems

Obstacle avoidance is an essential feature of autonomous robots operating in dynamic environments. Ultrasonic sensors are commonly used to measure distance and detect obstacles.

Many obstacle avoidance robots use ultrasonic sensors mounted on servo motors to scan the surrounding environment. The robot analyzes distance information and selects an alternative path when obstacles block the primary route.

Existing obstacle avoidance systems focus mainly on navigation and mobility. They do not usually incorporate plant security monitoring or real-time alert mechanisms.

The proposed Smart Plant Monitoring Bot combines obstacle avoidance with plant surveillance, enabling continuous operation even when obstacles appear on the patrol route.

2.5 Motion Detection Systems

Motion detection systems are commonly used in security and surveillance applications. Passive Infrared (PIR) sensors are widely used because of their low cost, low power consumption, and reliable performance.

A PIR sensor detects changes in infrared radiation emitted by moving objects. When movement is detected, the system can trigger alarms, notifications, or other protective actions.

Many home security systems utilize PIR sensors to monitor unauthorized movement. However, such systems are generally stationary and cannot patrol large areas.

The proposed Smart Plant Monitoring Bot integrates PIR-based motion detection with autonomous robotic movement, providing both surveillance and mobility.

2.6 GSM-Based Alert Systems

GSM technology enables remote communication through mobile networks. GSM modules are frequently used in security systems, industrial monitoring systems, and emergency alert systems.

When a specific event occurs, the GSM module sends SMS notifications to predefined phone numbers. This allows users to receive alerts even when they are away from the monitored location.

Several security systems use GSM communication to notify users about intrusion events. However, most of these systems are fixed installations without autonomous navigation capabilities.

The proposed system uses GSM communication to provide immediate notification whenever motion is detected near protected plants.

2.7 Comparative Analysis of Existing Systems

Table 2.1 compares the major features of existing systems with the proposed Smart Plant Monitoring Bot.

Table 2.1: Comparison of Existing Systems

The comparison shows that existing systems generally focus on a single functionality such as navigation, monitoring, or security. The proposed Smart Plant Monitoring Bot integrates all these features into a single autonomous platform.

Table 2.1: Comparison of Existing Systems with Proposed Smart Plant Monitoring Bot

Features	Smart Agriculture Monitoring	Line Following Robot	Security Alarm System	Proposed Smart Plant Monitoring Bot
Plant Monitoring	Yes	No	No	Yes
Autonomous Navigation	No	Yes	No	Yes
Obstacle Avoidance	Limited	Limited	No	Yes
Motion Detection	Limited	No	Yes	Yes
SMS Alert	Some Systems	No	Yes	Yes
Mobile Surveillance	No	No	No	Yes
Low Cost Implementation	Moderate	Yes	Yes	Yes

2.8 Research Gap

Based on the literature review, several limitations were identified in existing systems:

- Most plant monitoring systems are stationary.
- Many agricultural robots require expensive cameras and image processing systems.
- Conventional line-following robots do not provide plant protection functionality.
- Security monitoring systems lack autonomous mobility.
- Existing solutions often require internet connectivity for operation.

The proposed Smart Plant Monitoring Bot addresses these limitations by combining autonomous navigation, obstacle avoidance, motion detection, LCD monitoring, alarm generation, and GSM communication using low-cost hardware components.

2.9 Contribution of the Proposed System

The major contributions of the proposed system are:

1. Development of an autonomous robotic platform for plant protection.
2. Integration of line following and obstacle avoidance functionalities.
3. Real-time threat detection using PIR sensors.
4. LCD-based monitoring and status display.
5. GSM-based SMS alert generation.
6. Automatic resumption of patrol operation after threat removal.
7. Low-cost implementation suitable for academic and prototype applications.

2.10 Summary

This chapter reviewed existing smart agriculture systems, plant monitoring robots, line-following robots, obstacle avoidance systems, motion detection technologies, and GSM-based alert systems. The review identified several limitations in current solutions, particularly the lack of integration between autonomous navigation and plant protection. The proposed Smart Plant Monitoring Bot addresses these limitations by combining mobility, obstacle avoidance, threat detection, alarm generation, LCD monitoring, and GSM-based communication in a single robotic platform. The next chapter presents the system analysis and design of the proposed solution.

Chapter 3

System Analysis and Design

This chapter presents the analysis and design of the proposed Smart Plant Monitoring Bot. The chapter discusses the system requirements, overall architecture, hardware components, software requirements, working principles, and system design. The proposed system combines robotics, embedded systems, obstacle avoidance, motion detection, and GSM communication to provide autonomous plant monitoring and protection.

The robot continuously patrols along a predefined path using line-following technology. It detects obstacles, avoids collisions, monitors plant surroundings, and generates alerts whenever a potential threat is detected near protected plants.

3.1 System Requirements

The Smart Plant Monitoring Bot requires both hardware and software resources for successful implementation.

3.1.1 Hardware Requirements

The following hardware components are used in the project:

Table 3.1: Hardware Components Used in the Proposed System

Component	Quantity	Purpose
Arduino UNO	1	Main controller
L298N Motor Driver	1	Motor control
DC Motors	2	Robot movement
IR Line Tracking Sensors	2	Line following
HC-SR04 Ultrasonic Sensor	1	Obstacle detection
HC-SR501 PIR Sensor	1	Motion detection
Servo Motor	1	Sensor scanning
SIM800L GSM Module	1	SMS notification
16×2 I2C LCD	1	Status display
Buzzer	1	Audio alert
12V Power Supply	1	Power source
Robot Chassis	1	Physical structure

3.1.2 Software Requirements

The following software tools are used:

Table 3.2: Software Requirements

Software	Purpose
Arduino IDE	Code development
Arduino Libraries	Hardware interfacing
Fritzing	Circuit design
Microsoft Word / LaTeX	Documentation
GitHub	Version control

3.2 Functional Requirements

The Smart Plant Monitoring Bot must perform the following functions:

1. Follow a predefined black line path.
2. Detect obstacles during movement.
3. Stop when obstacles are detected.
4. Scan surroundings using ultrasonic sensors.
5. Select an alternative path around obstacles.
6. Search and return to the original line.
7. Detect movement near protected plants.
8. Generate buzzer alerts during threat detection.
9. Display warning messages on LCD.
10. Measure and display threat distance.
11. Send SMS alerts through GSM.
12. Resume patrol after threat removal.

3.3 Non-Functional Requirements

The system should satisfy the following requirements:

Reliability

The robot should perform continuous monitoring without frequent failures.

Accuracy

The sensors should provide accurate line detection, obstacle detection, and motion detection.

Cost Efficiency

The system should use affordable components suitable for academic projects.

Maintainability

The hardware and software should be easy to modify and upgrade.

Safety

The robot should avoid collisions and operate safely around plants.

3.4 System Architecture

The Smart Plant Monitoring Bot consists of sensing, processing, communication, and actuation units.

The Arduino UNO acts as the central controller. Input data from the IR sensors, PIR sensor, and ultrasonic sensor are processed by the controller. Based on the decision-making algorithm, the Arduino controls the motors, servo motor, buzzer, LCD display, and GSM module.

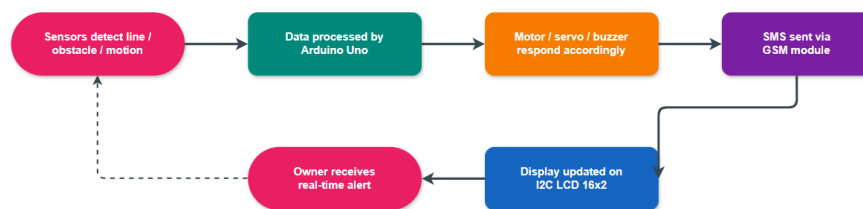


Figure 3.1: Overall System Architecture

The architecture consists of:

Input Layer

- IR Sensors.
- Ultrasonic Sensor.
- PIR Sensor.

Processing Layer

- Arduino UNO

Output Layer

- DC Motors.
- Servo Motor.
- LCD Display .
- Buzzer .
- GSM Module.

3.5 Working Principle

The robot continuously patrols the plant monitoring area using line-following technology.

The IR sensors detect the black guiding line and control robot navigation. During movement, the ultrasonic sensor continuously measures the distance of objects in front of the robot.

If an obstacle is detected within a predefined distance, the robot stops and scans both left and right directions using a servo-mounted ultrasonic sensor. The robot selects the clearer path, bypasses the obstacle, searches for the line, and resumes normal operation.

Simultaneously, the PIR sensor monitors movement around protected plants. When motion is detected, the robot immediately stops patrolling and enters threat monitoring mode. The buzzer is activated, the LCD displays warning messages, and an SMS alert is sent through the GSM module. Once the threat leaves the monitored area, the robot resumes normal patrol operation.

3.6 Block Diagram

The block diagram illustrates the interaction among various system modules.

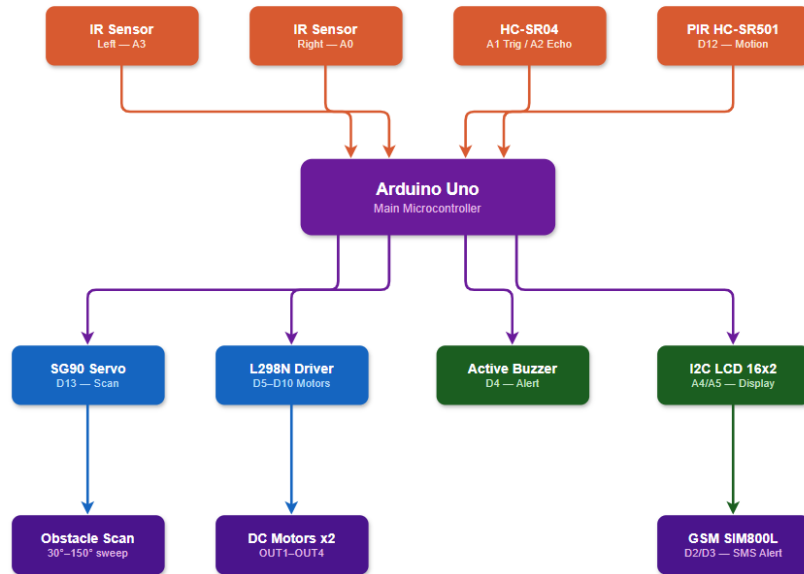


Figure 3.2: Embedded System Diagram of Smart Plant Monitoring Bot

3.7 Circuit Design

The circuit integrates all sensors and output devices with Arduino UNO.

The major connections include:

IR Sensors

- Connected to digital input pins.
- Used for line tracking.

Ultrasonic Sensor

- Trigger pin connected to Arduino output.
- Echo pin connected to Arduino input.

PIR Sensor

- Output connected to digital input pin.

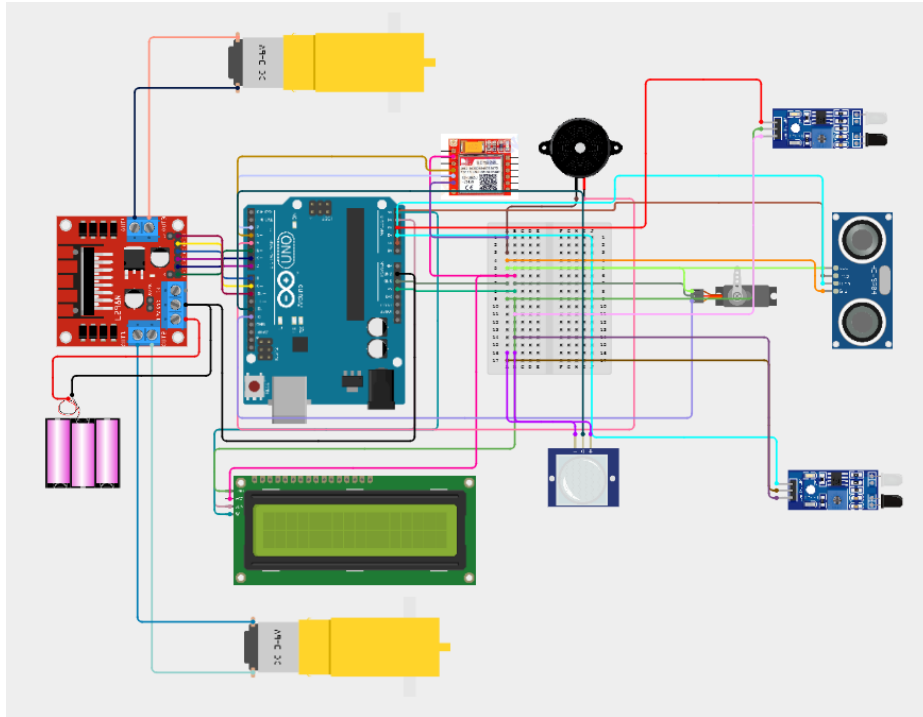


Figure 3.3: Circuit Diagram of Smart Plant Monitoring Bot

Servo Motor

- Signal pin connected to PWM output pin.

GSM Module

- Connected using serial communication.

LCD Display

- Connected through I2C interface.

Buzzer

- Connected to digital output pin.

Motor Driver

- Connected between Arduino and DC motors.

3.8 Line Following Design

The navigation system is based on two IR sensors.

Table 3.3: Line Following Logic

Left IR	Right IR	Action
Black	Black	Move Forward
Black	White	Turn Left
White	Black	Turn Right
White	White	Search Line

The robot continuously adjusts its direction according to sensor readings.

3.9 Obstacle Avoidance Design

The obstacle avoidance subsystem uses the ultrasonic sensor and servo motor. The process includes:

1. Detect obstacle.
2. Stop movement.
3. Beep twice.
4. Display obstacle warning.
5. Scan left side.
6. Scan right side.
7. Compare distances.
8. Select clearer path.
9. Bypass obstacle.
10. Search line.
11. Resume patrol.

3.10 Threat Detection

The threat detection subsystem protects plants from unwanted disturbances. When motion is detected:

1. Robot stops.
2. PIR confirms movement.
3. Buzzer activates continuously.
4. LCD displays warning.
5. Distance is measured.
6. SMS notification is sent.
7. Monitoring continues.
8. Patrol resumes after threat removal.

3.11 Activity Diagram

3.12 Summary

This chapter presented the system analysis and design of the Smart Plant Monitoring Bot. The chapter discussed hardware and software requirements, system architecture, circuit design, block diagram, line-following logic, obstacle avoidance mechanism, threat detection process, and overall system workflow. The proposed design integrates autonomous navigation, obstacle avoidance, motion detection, alert generation, and GSM communication into a single robotic platform. The next chapter describes the methodology and implementation of the system.

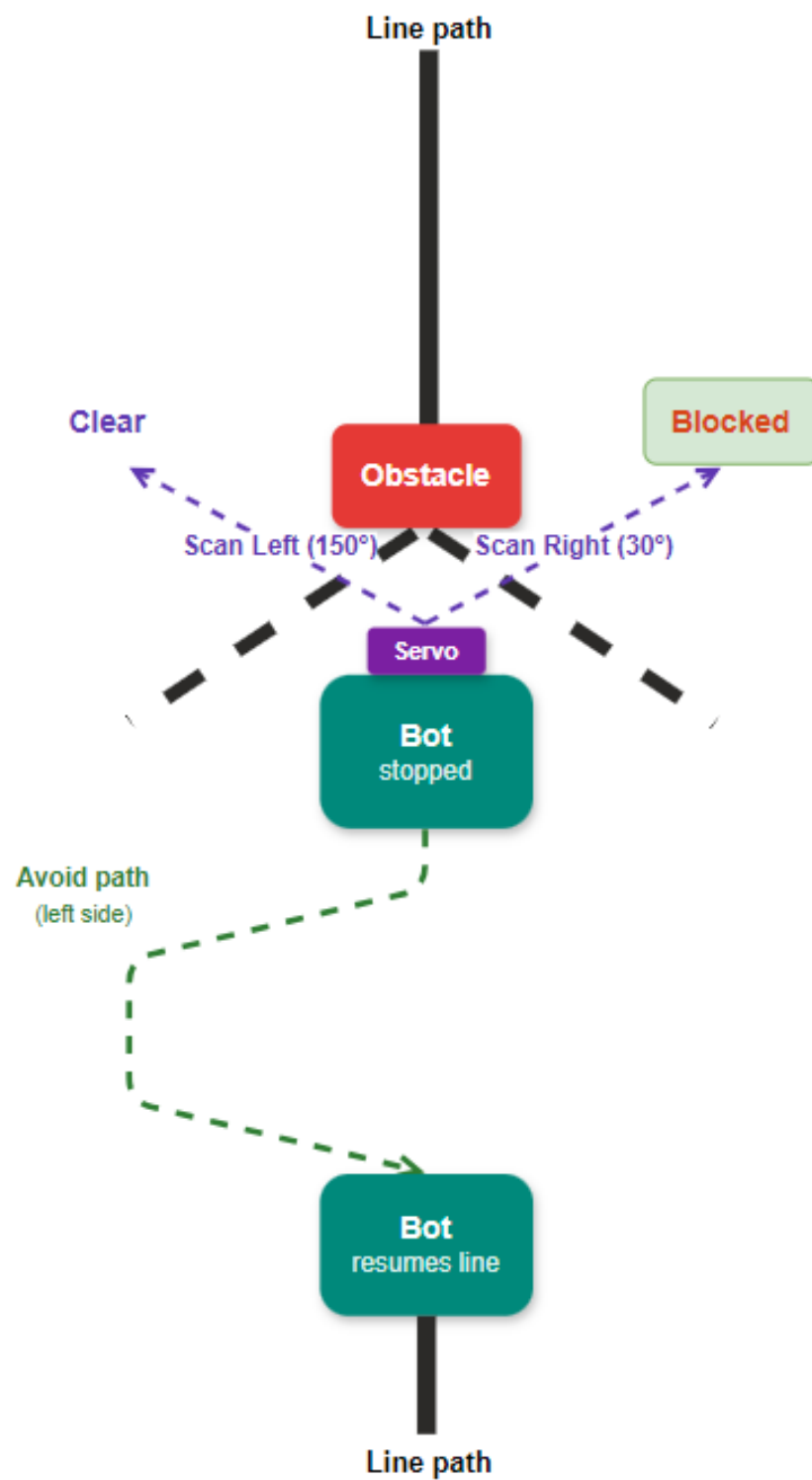


Figure 3.4: Obstacle Avoidance Mechanism

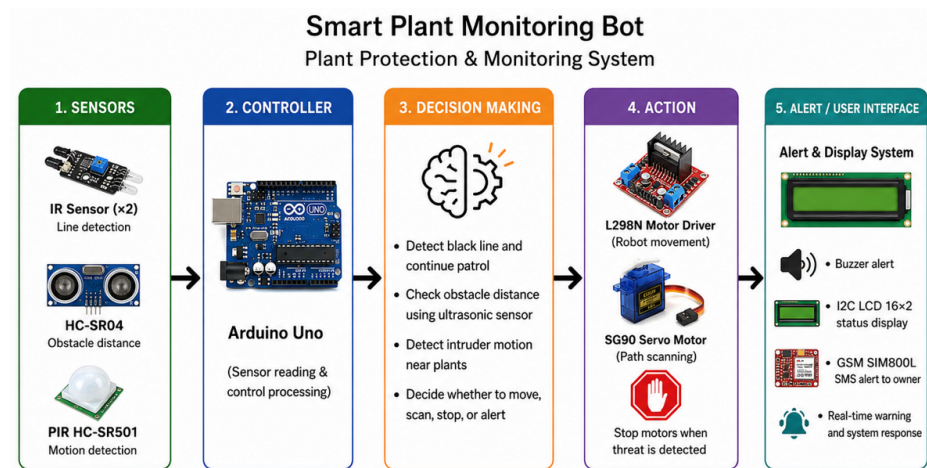


Figure 3.5: Activity Diagram

Chapter 4

Methodology and Implementation

This chapter presents the methodology and implementation of the Smart Plant Monitoring Bot. It describes the development approach, hardware integration, software implementation, system workflow, algorithms, and testing procedures used to develop the prototype.

The project combines robotics, embedded systems, sensor technology, and GSM communication to create an autonomous plant monitoring and protection system.

4.1 Development Methodology

The project was developed using a prototype-based methodology. The development process included requirement analysis, component selection, system design, hardware assembly, software implementation, testing, and validation. Development Steps:

1. Requirement Analysis
2. Component Selection
3. System Design
4. Circuit Development
5. Hardware Assembly
6. Software Development

7. Testing and Debugging

8. Final Integration

9. Performance Evaluation

4.2 Hardware Implementation

The hardware implementation consists of sensors, communication modules, motor control units, and display devices connected to the Arduino UNO.

Main Hardware Modules:

Arduino UNO

Acts as the central processing unit.

IR Sensors

Used for line detection and navigation.

Ultrasonic Sensor Used for obstacle detection and distance measurement.

PIR Sensor Used for motion detection around plants.

Servo Motor

Rotates the ultrasonic sensor for environmental scanning.

GSM Module

Sends SMS notifications to the user.

LCD Display

Displays system status and alerts.

Buzzer

Generates audible warning signals.

298N Motor Driver

Controls DC motor movement.

DC Motors Provide mobility to the robot.

4.3 Hardware Assembly

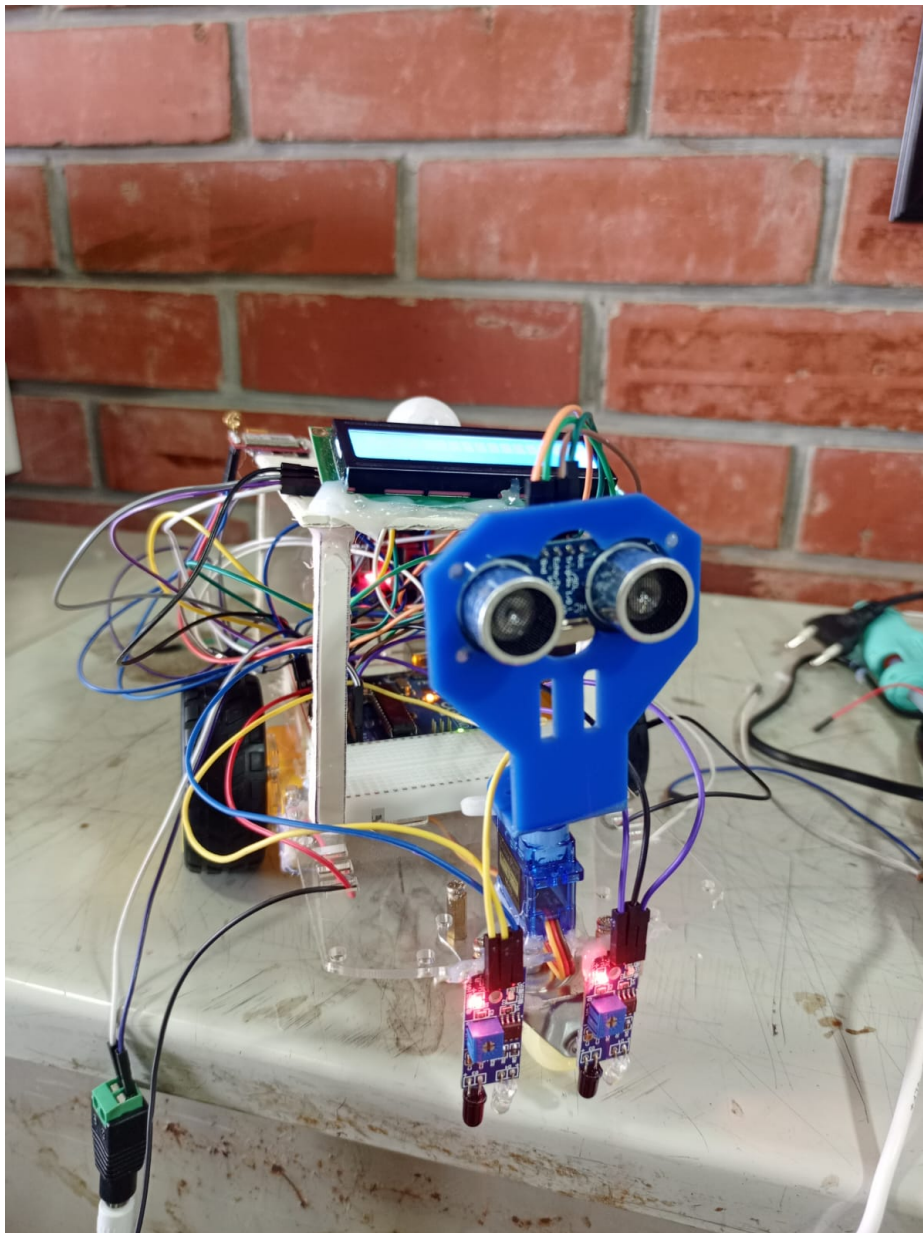


Figure 4.1: Hardware Setup

4.4 Software Implementation

The software was developed using Arduino IDE and Arduino C/C++ programming language.

The program consists of several functional modules:

Line Following Module Responsible for path navigation using IR sensor data.

Obstacle Avoidance Module Responsible for detecting and bypassing obstacles.

Motion Detection Module Responsible for identifying threats around plants.

Alert Management Module Controls buzzer, LCD messages, and GSM communication.

Navigation Recovery Module Searches and returns to the line after obstacle avoidance.

4.5 System Workflow

The robot continuously executes a sequence of monitoring and navigation operations.

Overall Workflow

- Start system.
- Initialize sensors and modules.
- Follow black line.
- Check obstacle status..
- Check motion status.
- Generate alerts if required.
- Resume patrol operation.

This section focuses on how the system was implemented — not on design specifications already covered earlier.

4.6 Overall Flowchart

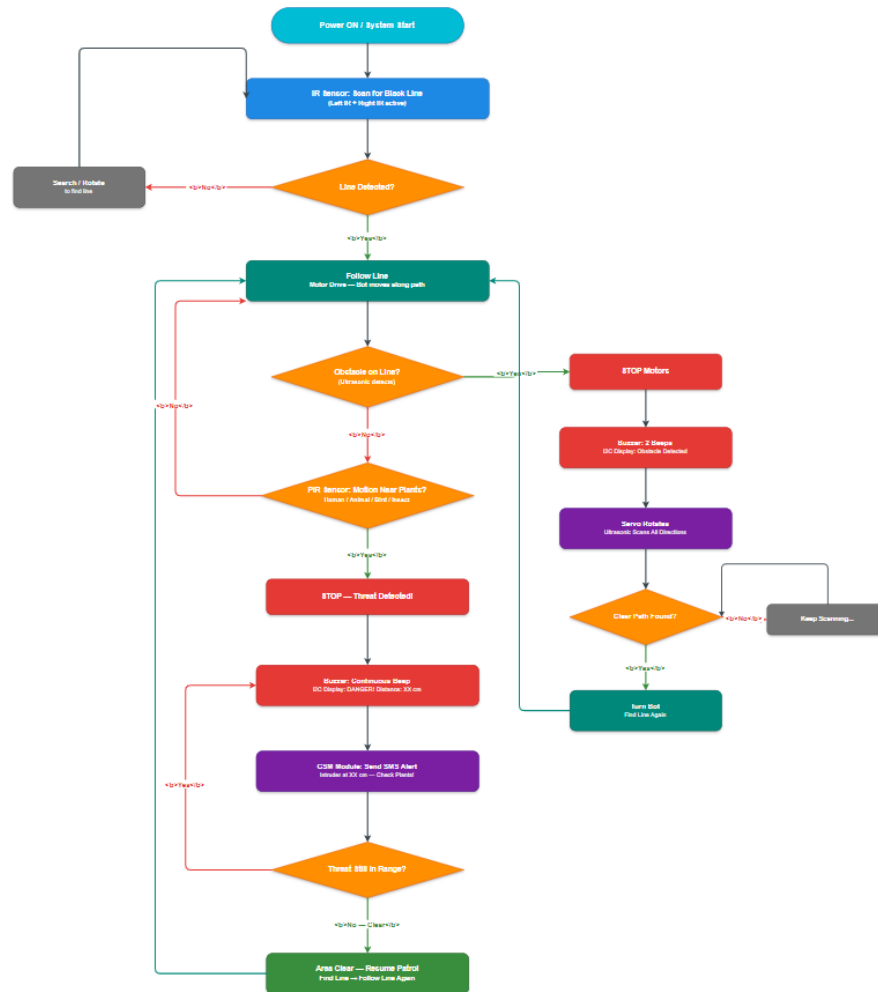


Figure 4.2: System Flow Diagram

4.7 Algorithm for Line Following

The robot uses two IR sensors to detect the path. **Algorithm 1: Line Following**

1. Read left IR sensor.

2. Read right IR sensor.
3. If both sensors detect black:
 - Move forward.
4. If left sensor detects black:
 - Turn left.
5. If right sensor detects black:
 - Turn right.
6. If both sensors lose the line:
 - Search for line.
7. Repeat continuously.

Table 4.1: Line Following Decision Table

Left IR	Right IR	Action
Black	Black	Forward
Black	White	Left
White	Black	Right
White	White	Search Line

4.8 Algorithm for Obstacle Avoidance

Obstacle avoidance is performed using the ultrasonic sensor and servo motor. **Algorithm 2: Obstacle Avoidance**

1. Measure front distance.
2. If distance less than threshold:
 - Stop robot.
 - Activate buzzer twice.

- Display obstacle warning.
- 3. Rotate servo left.
- 4. Measure left distance.
- 5. Rotate servo right.
- 6. Measure right distance.
- 7. Compare distances.
- 8. Select clearer direction.
- 9. Move around obstacle.
- 10. Search line.
- 11. Resume patrol.

4.9 Algorithm for Threat Detection

Threat detection protects plants from animals, birds, insects, and humans.

Algorithm 3: Threat Detection

- Read PIR sensor.
- If motion detected:
 - Stop robot.
 - Activate buzzer.
 - Measure distance.
 - Display warning on LCD.
 - Send SMS alert.
 - Continue monitoring.
- If threat leaves:

- Stop buzzer.
- Resume patrol.
- Return to line-following mode.

4.10 GSM Communication Implementation

The GSM module is responsible for notifying the user whenever a threat is detected.

SMS Alert Procedure

- Motion detected.
- Distance measured.
- GSM module activated.
- SMS prepared.
- SMS sent to registered number.

Example Message:

```
ALERT!  
Motion Detected Near Plants.  
Distance: 35 cm  
Smart Plant Monitoring Bot
```

4.11 LCD Display Implementation

The 16×2 I2C LCD provides real-time system information.

Normal Mode

```
SMART PLANT BOT  
Patrolling...
```



Figure 4.3: I2c Display reading

Obstacle Detection

Obstacle Found

Scanning Path

Danger Detection

DANGER ALERT

Dist: 35 cm

Table 4.2: LCD Messages

Event	LCD Message
Startup	System Ready
Patrol	Patrolling
Obstacle	Obstacle Found
Scanning	Scanning Path
Threat	Danger Alert
Resume	Patrol Active

4.12 Buzzer Implementation

The buzzer provides audible feedback.

Obstacle Detection

- Beep two times.

Threat Detection

- Continuous alarm until threat leaves.

Table 4.3: Buzzer Operations

Condition	Buzzer Action
Startup	Single Beep
Obstacle	Two Beeps
Threat	Continuous Alarm

4.13 Pin Configuration

(Modify according to your actual wiring.)

4.14 Program Structure

The software is divided into several functions.

Table 4.4: Arduino Pin Mapping

Component	Arduino Pin
Left IR Sensor	D2
Right IR Sensor	D3
Ultrasonic Trigger	D4
Ultrasonic Echo	D5
PIR Sensor	D6
Servo Motor	D9
Buzzer	D10
Motor Driver IN1	D7
Motor Driver IN2	D8
Motor Driver IN3	D11
Motor Driver IN4	D12
GSM Module TX/RX	Software Serial
LCD SDA	A4
LCD SCL	A5

Main Functions

- `setup()`
- `loop()`
- `lineFollow()`
- `obstacleAvoid()`
- `findLineAfterAvoid()`
- `detectThreat()`
- `sendSMS()`
- `getDistance()`
- `moveForward()`
- `turnLeft()`
- `turnRight()`

- `stopRobot()`

This modular structure improves readability and simplifies debugging.

4.15 Testing and Debugging

Individual modules were tested before final integration.

Sensor Testing

- IR sensor response verified.
- Ultrasonic distance measurements checked.
- PIR detection range tested.

Communication Testing

- GSM SMS transmission verified.

Motor Testing

- Motor direction checked.
- Obstacle avoidance movement validated.

Integration Testing

- Complete system tested under real operating conditions.

4.16 Summary

This chapter presented the methodology and implementation of the Smart Plant Monitoring Bot. The development process, hardware integration, software architecture,

algorithms, GSM communication, LCD implementation, pin configuration, and testing procedures were discussed. The implemented system successfully combines autonomous navigation, obstacle avoidance, motion detection, and alert generation into a single robotic platform. The next chapter presents the experimental results, performance evaluation, discussion, and limitations of the proposed system.

Chapter 5

Results and Evaluation

This chapter presents the experimental results, performance evaluation, testing procedures, discussion, and limitations of the Smart Plant Monitoring Bot. The system was tested under various operating conditions to verify its functionality, reliability, and performance.

The evaluation focused on line-following performance, obstacle avoidance capability, threat detection accuracy, LCD display operation, buzzer functionality, and GSM-based SMS alert transmission.

5.1 Experimental Setup

The prototype was tested in a controlled indoor environment using a custom-designed black-line track. Several artificial obstacles were placed on the track to evaluate obstacle avoidance performance. Human movement was used to simulate threats near the protected plants.

The experimental setup consisted of:

- Smart Plant Monitoring Bot
- Black-line navigation track
- Artificial obstacles

- Protected plant area
- GSM communication network
- LCD monitoring system

5.2 Functional Testing

Each subsystem was tested individually before evaluating the complete integrated system.

5.2.1 Line Following Test

The robot was placed on the predefined track and allowed to navigate autonomously.

Test Objective

To verify whether the robot could correctly follow the black guiding line.

Result

The robot successfully followed the path and adjusted its direction according to sensor readings.

Table 5.1: Line Following Test Results

Test Condition	Expected Result	Actual Result	Status
Straight Path	Move Forward	Successful	Pass
Left Turn	Turn Left	Successful	Pass
Right Turn	Turn Right	Successful	Pass
Sharp Curve	Follow Curve	Successful	Pass
Line Loss	Search Line	Successful	Pass

Figure 5.1: Line Following Test

5.2.2 Obstacle Detection and Avoidance Test

Obstacles were placed on the robot path.

Test Objective

To verify obstacle detection and avoidance capability.

Procedure

1. Place obstacle on track.
2. Allow robot to approach.
3. Observe avoidance behavior.

Result

The robot detected obstacles successfully and selected an alternative route using servo-based environmental scanning.

Table 5.2: Obstacle Avoidance Test Results

Test	Obstacle De- tected	Path Scanned	Obstacle Avoided	Status
Test 1	Yes	Yes	Yes	Pass
Test 2	Yes	Yes	Yes	Pass
Test 3	Yes	Yes	Yes	Pass
Test 4	Yes	Yes	Yes	Pass

Figure 5.2: Obstacle Avoidance Test

5.2.3 PIR Motion Detection Test

Motion was generated near the protected plants.

Test Objective

To verify threat detection capability.

Procedure

1. Activate patrol mode.
2. Move near the plant area.
3. Observe system response.

Result

The PIR sensor successfully detected movement and activated the alert system.

Table 5.3: PIR Motion Detection Results

Trial	Motion Detected	De-	Alarm Activated	LCD Updated	Status
1	Yes		Yes	Yes	Pass
2	Yes		Yes	Yes	Pass
3	Yes		Yes	Yes	Pass
4	Yes		Yes	Yes	Pass

Figure 5.3: Motion Detection Test

5.2.4 GSM Alert Testing

The GSM module was tested to verify SMS delivery.

Test Objective

To verify remote alert generation.

Procedure

1. Trigger PIR sensor.
2. Generate danger condition.

3. Monitor SMS transmission.

Result

SMS notifications were successfully delivered to the registered mobile number.

Example SMS

ALERT!

Motion Detected Near Plants.

Distance: 30 cm

Smart Plant Monitoring Bot

Table 5.4: GSM Alert Results

Trial	SMS Sent	SMS Received	Status
1	Yes	Yes	Pass
2	Yes	Yes	Pass
3	Yes	Yes	Pass
4	Yes	Yes	Pass

Figure 5.4: GSM Alert Test

5.2.5 LCD Display Testing

The LCD was tested under different operating conditions.

Table 5.5: LCD Display Results

Event	Message Displayed	Status
Startup	System Ready	Pass
Patrol Mode	Patrolling	Pass
Obstacle Detection	Obstacle Found	Pass
Path Scanning	Scanning Path	Pass
Threat Detection	Danger Alert	Pass
Resume Patrol	Patrol Active	Pass

Figure 5.5: LCD Display Results

5.2.6 Buzzer Testing

The buzzer functionality was verified.

Table 5.6: Buzzer Test Results

Condition	Expected Action	Actual Action	Status
Startup	Single Beep	Correct	Pass
Obstacle	Two Beeps	Correct	Pass
Threat	Continuous Alarm	Correct	Pass

5.3 Performance Summary

The Smart Plant Monitoring Bot successfully completed all primary tasks.

Table 5.7: Performance Summary

Feature	Performance
Line Following	Successful
Obstacle Detection	Successful
Obstacle Avoidance	Successful
PIR Detection	Successful
GSM Alert	Successful
LCD Display	Successful
Buzzer Alert	Successful
Patrol Recovery	Successful

The integrated system operated reliably during testing and demonstrated stable performance throughout the evaluation period.

5.4 Discussion on Findings

The experimental results indicate that the primary objectives of the project were successfully achieved.

The robot demonstrated reliable line-following performance and was able to navigate the predefined route without significant deviation. The IR sensors provided accurate path detection and helped maintain stable movement.

The ultrasonic sensor effectively detected obstacles and measured distances. The servo motor enhanced obstacle avoidance performance by allowing environmental scanning in multiple directions. This enabled the robot to select a safer path and continue patrolling without manual intervention.

The PIR sensor successfully detected motion near protected plants. Upon detection, the robot immediately stopped movement, activated the buzzer alarm, displayed warning information on the LCD, and sent SMS notifications through the GSM module.

The GSM communication system provided an effective remote alert mechanism. Users could receive immediate notification even when they were away from the monitored area.

Compared with manual plant monitoring methods, the proposed system reduces human effort and improves surveillance efficiency. The integration of robotics, obstacle avoidance, and security monitoring makes the system suitable for smart agriculture and automated plant protection applications.

5.5 Limitations

Although the prototype performed successfully, several limitations were identified.

Limited Detection Capability

The PIR sensor can detect motion but cannot identify the specific type of object. It cannot distinguish between humans, birds, cats, dogs, or insects.

Small-Scale Prototype

The system was tested on a laboratory-scale setup. Real agricultural deployment may require larger chassis, stronger motors, and improved durability.

GSM Dependency

SMS notification depends on GSM network availability and signal strength.

Environmental Conditions

IR sensors may be affected by extreme lighting conditions, while ultrasonic measurements can be influenced by irregular object surfaces.

Power Consumption

Long-term outdoor deployment would require improved battery management and power optimization.

5.6 Summary

This chapter presented the experimental setup, testing procedures, performance evaluation, findings, and limitations of the Smart Plant Monitoring Bot. The results demonstrate successful line following, obstacle avoidance, motion detection, LCD monitoring, buzzer alert generation, and GSM-based SMS notification. Overall, the project successfully achieved its objectives and demonstrated a practical application of robotics and embedded systems for smart plant monitoring and protection.

Chapter 6

Project Planning and Budget

Proper planning and resource management are essential for the successful completion of any engineering project. This chapter presents the project schedule, work breakdown structure, resource allocation, and budget estimation of the Smart Plant Monitoring Bot.

The project was completed through several stages including requirement analysis, system design, hardware assembly, software development, testing, and documentation. Effective planning ensured that all project objectives were achieved within the available time and resources.

6.1 Project Planning

The development process was divided into multiple phases to ensure systematic implementation and testing.

Major Development Phases

- Requirement Analysis
- Literature Review
- System Design
- Hardware Collection

- Circuit Development
- Programming
- Hardware Integration
- Testing and Debugging
- Final Evaluation
- Report Preparation

6.2 Project Timeline

The project was planned over a period of 12 weeks.

Table 6.1: Project Schedule

Activities	1	2	3	4	5	6	7	8	9	10	11	12
Requirement Analysis												
Literature Review												
System Design												
Hardware Collection												
Circuit Design												
Programming												
Hardware Assembly												
Integration												
Testing												
Debugging												
Evaluation												
Documentation												
Final Submission												

Gantt Chart

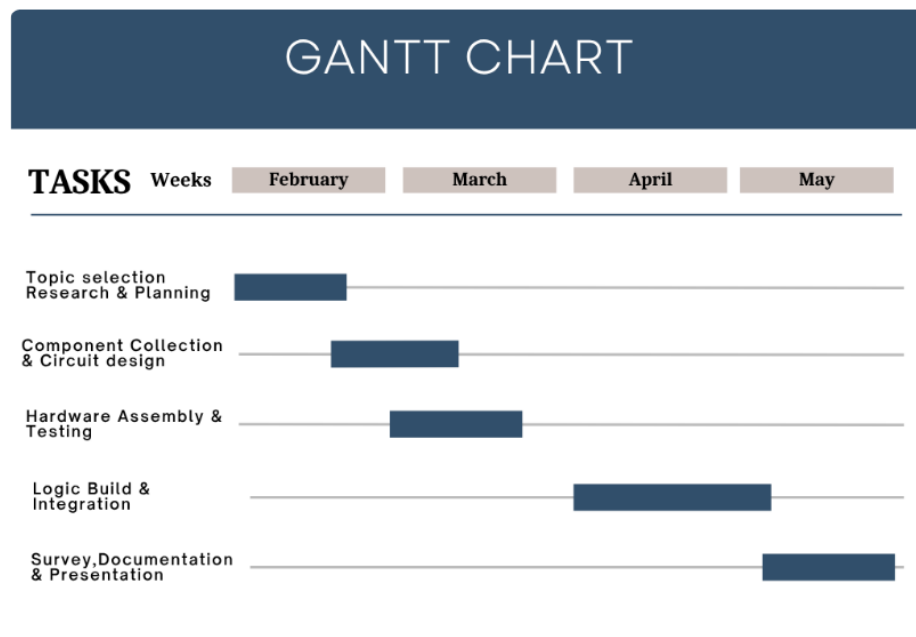


Figure 6.1: Gantt Chart

6.3 Resource Planning

Several hardware and software resources were required during project development.

Hardware Resources

- Arduino UNO
- L298N Motor Driver
- DC Motors
- IR Sensors
- Ultrasonic Sensor
- PIR Sensor
- Servo Motor
- GSM Module

- LCD Display
- Buzzer
- Power Supply
- Robot Chassis

Software Resources

- Arduino IDE
- Fritzing
- Microsoft Office
- GitHub

6.4 Budget Estimation

The budget estimation was prepared based on local market prices of electronic components.

6.5 Software Cost Estimation

The software tools used in this project are open-source or freely available.

6.6 Risk Management

Several risks were identified during project development.

Table 6.2: Hardware Cost Estimation

Component	Quantity	Unit Cost (BDT)	Total Cost (BDT)
Arduino UNO	1	650	650
L298N Motor Driver	1	250	250
DC Motors & Chassis	1	600	600
IR Sensors	2	80	160
Ultrasonic Sensor HC-SR04	1	100	100
PIR Sensor HC-SR501	1	100	100
Servo Motor SG90	1	150	150
GSM Module SIM800L	1	200	200
LCD 16×2 I2C	1	200	200
Buzzer	1	30	30
Battery/Power Supply	1	200	200
Jumper Wires	2 Set	200	200
Breadboard/PCB	1	100	100
Total Estimated Hardware Cost			2,940

6.7 Project Deliverables

The final deliverables of the project include:

Hardware Deliverables

- Smart Plant Monitoring Bot Prototype
- Integrated Sensor System
- GSM Communication System

Software Deliverables

- Arduino Source Code
- Flowcharts
- Circuit Design

Table 6.3: Software Cost

Software	Cost
Arduino IDE	Free
Fritzing	Free/Educational
GitHub	Free
Microsoft Office	Available through Institution
Total Software Cost	BDT = 0

Table 6.4: Risk Assessment

Risk	Impact	Mitigation Strategy
Sensor Failure	Medium	Component Testing
GSM Network Issue	Medium	Signal Verification
Battery Drain	Medium	Proper Power Management
Wiring Errors	High	Circuit Verification
Software Bugs	High	Continuous Testing
Motor Failure	Medium	Spare Components

Documentation Deliverables

- Project Report
- Presentation Slides
- Testing Results

6.8 Sustainability and Future Expansion Planning

The project is designed using low-cost and easily available electronic components. Future versions can incorporate:

- Solar Power Integration
- ESP32-Based IoT Monitoring
- Mobile Application Control

- Camera-Based Threat Detection
- AI-Based Animal Recognition
- Cloud Data Logging

These improvements can significantly increase the functionality and scalability of the system.

6.9 Summary

This chapter presented the project planning, work breakdown structure, project schedule, resource allocation, budget estimation, risk assessment, and expected deliverables of the Smart Plant Monitoring Bot. The estimated project cost is approximately BDT 2,940, making the system affordable for academic and prototype-level applications. Proper planning and resource management contributed to the successful completion of the project. The next chapter discusses engineering standards, design constraints, CEP mapping, CEA mapping, and knowledge profile requirements associated with the project.

Chapter 7

Engineering Standards, Design Constraints and CEP Mapping

Engineering projects must be developed following appropriate standards, constraints, ethical considerations, and professional engineering practices. This chapter discusses the engineering standards, design constraints, sustainability considerations, Complex Engineering Activities (CEA), Complex Engineering Problems (CEP), and Knowledge Profile mapping associated with the Smart Plant Monitoring Bot.

The project integrates robotics, embedded systems, sensor technologies, communication systems, and automation to address plant monitoring and protection challenges.

7.1 Engineering Standards

Several engineering standards and best practices were considered during the design and implementation of the Smart Plant Monitoring Bot. [1–4]

7.1.1 Electrical Safety Standards

Electrical safety was maintained throughout the system design. Key considerations included:

- Proper voltage regulation.

- Safe wiring connections.
- Short-circuit prevention.
- Controlled power distribution.
- Protection of electronic components.

7.1.2 Embedded System Standards

The software development process followed common embedded system design principles:

- Modular programming.
- Real-time response handling.
- Sensor calibration.
- Error reduction techniques.
- Reliable communication protocols.

7.1.3 Communication Standards

The GSM communication module follows standard mobile communication protocols. The system:

- Uses AT Commands.
- Supports SMS transmission.
- Provides remote alert capability.
- Ensures reliable message delivery.

7.1.4 Robotics and Automation Standards

The robot follows fundamental automation principles:

- Autonomous operation.
- Sensor-based decision making.
- Obstacle avoidance.
- Real-time environmental monitoring.
- Automatic recovery and continuation of patrol.

7.2 Design Constraints

Several constraints were considered during project development.

7.2.1 Economic Constraints

The project was designed using low-cost components. Examples:

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- PIR Sensor
- SIM800L GSM Module

The total project cost was kept within a reasonable academic budget.

7.2.2 Technical Constraints

The system has several technical limitations:

- PIR sensor cannot identify object types.
- Ultrasonic sensor performance depends on object surfaces.
- GSM communication depends on network availability.
- IR sensors may be affected by lighting conditions.

7.2.3 Environmental Constraints

Outdoor deployment may face challenges such as:

- Rain
- Dust
- High temperature
- Humidity
- Uneven terrain

Additional protective housing may be required for practical deployment.

7.2.4 Operational Constraints

The robot operates on a predefined path. Therefore:

- Continuous line availability is required.
- Battery charging is necessary.
- Sensor maintenance may be required periodically.

7.3 Sustainability Considerations

The Smart Plant Monitoring Bot supports sustainable agricultural practices. Benefits include:

- Reduced plant damage.
- Reduced manual supervision.
- Efficient resource utilization.
- Low power consumption.
- Reusable electronic components.

The project demonstrates how automation can contribute to sustainable farming practices.

7.4 Ethical and Professional Considerations

The project follows ethical engineering practices. Important considerations include:

- Responsible use of GSM communication.
- Accurate reporting of results.
- Safe system operation.
- Environmental responsibility.
- Respect for user privacy.

The system only monitors movement and does not collect personal information.

7.5 Health and Safety Considerations

Several safety measures were considered.

Electrical Safety

- Proper insulation.
- Secure connections.
- Safe voltage levels.

Mechanical Safety

- Stable robot structure.
- Controlled motor operation.
- Safe servo movement.

User Safety

- Low-voltage operation.
- Non-hazardous components.
- Audible warning system.

7.6 Complex Engineering Problem (CEP) Analysis

The Smart Plant Monitoring Bot addresses a complex engineering problem involving autonomous navigation, obstacle avoidance, real-time monitoring, sensor integration, and wireless communication. The project requires the integration of multiple engineering domains including embedded systems, robotics, electronics, programming, and communication technologies.

7.7 Risk and Reliability Considerations

Potential risks include:

Sensor Failure

May affect obstacle detection or threat monitoring.

Communication Failure

May delay SMS notifications.

Power Failure

May interrupt patrol operation.

Software Errors

May affect system decisions.

To reduce risks:

- Regular testing was performed.
- Modular programming was used.
- Sensor calibration was conducted.
- Error handling procedures were implemented.

7.8 Summary

This chapter discussed the engineering standards, design constraints, sustainability considerations, ethical issues, health and safety aspects, CEP analysis, CEA analysis, and Knowledge Profile mapping of the Smart Plant Monitoring Bot. The project successfully satisfies the requirements of complex engineering problems and activities through the integration of robotics, embedded systems, sensor technologies, and com-

munication systems. The next chapter presents the conclusion and future work of the project.

Table 7.1: Mapping with Complex Engineering Problems (Ps)

Ps	Attributes	How Ps are addressed	Evidence
P1	Depth of Knowledge	Applies embedded systems, sensors, motor control, ultrasonic ranging, GSM communication, and servo actuation.	Ch. 2, 3, 4
P2	Conflicting Requirements	Balances low cost, real-time line tracking, smooth obstacle avoidance, and reliable SMS alerting.	Ch. 3, 5
P3	Depth of Analysis	Analyzes IR sensor response, ultrasonic distance calculation, PIR motion behavior, and servo scan logic.	Ch. 4, 5
P4	Familiarity of Issues	Line following is known, but combining obstacle avoidance with PIR-based intruder alert creates project-specific challenges.	Ch. 1, 2, 3
P5	Applicable Standards	Uses standard Arduino libraries, L298N motor driver interfacing, and GSM AT command practices.	Sec. 7.1
P6	Stakeholder Needs	Supports home plant owners, gardeners, and small farm caretakers through autonomous monitoring.	Ch. 3, Appendix
P7	Interdependence	IR sensors, ultrasonic, servo, PIR, buzzer, LCD, and GSM module are interconnected; failure of one module affects the system.	Ch. 3, 4, 5

Table 7.2: Mapping of Complex Engineering Activities (As)

As	Attributes	How As are addressed	Evidence
A1	Range of Resources	Uses Arduino Uno, IR sensors, ultrasonic sensor, servo motor, PIR sensor, GSM module, LCD, L298N driver, and libraries.	Ch. 3, 4
A2	Level of Interaction	Multiple modules interact, including IR sensors, ultrasonic, servo, PIR, buzzer, LCD, and GSM module.	Ch. 3, 4
A3	Innovation	Uses a servo-mounted ultrasonic sensor for dynamic obstacle scanning instead of a fixed sensor.	Ch. 3, 5
A4	Society and Environment	Can reduce plant damage, support home gardening, and reduce manual monitoring effort.	Ch. 1, 7
A5	Familiarity	Combines known technologies into a practical robotics-based plant protection prototype.	Ch. 2, 4

Table 7.3: Knowledge Profile Mapping (Ks)

Profile	Knowledge Area	Project Evidence
K1	Natural science and engineering fundamentals	Sensor detection, electrical interfacing, and motor control.
K2	Mathematics and computing	Distance calculation logic, timing control, and conditional decision making.
K3	Engineering fundamentals	Embedded system design using Arduino and electronic modules.
K4	Specialist knowledge	Ultrasonic ranging, PIR motion detection, and GSM-based SMS alerting.
K5	Engineering design	System architecture, circuit design, and hardware-software integration.
K6	Engineering practice	Prototype development, testing, debugging, and documentation.
K7	Social, environmental, and safety knowledge	Plant protection, safety, and sustainable low-power design considerations.
K8	Research literature and standards	Related works review and use of technical documentation.

Chapter 8

Conclusion and Future Work

This chapter presents the conclusion of the Smart Plant Monitoring Bot project and discusses possible future improvements. The project successfully integrates robotics, embedded systems, sensor technologies, and wireless communication to develop an autonomous plant monitoring and protection system.

The system was designed to monitor plants continuously, detect obstacles during navigation, identify nearby threats through motion detection, generate warning alerts, and notify users remotely through GSM communication.

8.1 Project Summary

The Smart Plant Monitoring Bot was developed to address the challenges associated with continuous plant monitoring and protection. Traditional monitoring systems often require human supervision and are inefficient for long-term observation.

To overcome these limitations, an autonomous robotic platform was designed using:

- Arduino UNO
- IR Line Tracking Sensors
- HC-SR04 Ultrasonic Sensor
- HC-SR501 PIR Sensor

- Servo Motor
- SIM800L GSM Module
- 16×2 I2C LCD Display
- Buzzer
- L298N Motor Driver
- DC Motors

The robot patrols along a predefined path using line-following technology while continuously monitoring the surrounding environment. When an obstacle is detected, the robot automatically scans the surrounding area and selects an alternative route. If motion is detected near protected plants, the robot immediately activates an alarm, displays warning information, measures distance, and sends SMS notifications to the owner.

The developed prototype demonstrates the practical application of robotics and embedded systems in smart agriculture and automated plant protection.

8.2 Advantages of the Proposed System

The developed system offers several advantages.

Reduced Human Effort

Continuous monitoring can be performed automatically.

Increased Plant Protection

The system provides early warning against potential threats.

Real-Time Notification

SMS alerts allow users to respond quickly.

Autonomous Navigation

No manual movement control is required.

Low Cost

The project uses inexpensive components suitable for educational and prototype applications.

Expandability

Additional sensors and IoT technologies can easily be integrated in future versions.

8.3 Limitations of the Project

Although the project achieved its objectives, several limitations remain.

PIR Sensor Limitation

The PIR sensor detects motion but cannot distinguish whether the object is a human, bird, dog, cat, or insect.

GSM Dependency

SMS notifications depend on mobile network availability.

Limited Coverage Area

The robot can only monitor areas covered by the predefined track.

Environmental Influence

Sensor performance may be affected by environmental conditions.

Battery Dependency

Continuous operation depends on available battery power.

Prototype Scale

The project was developed as a small-scale prototype and may require additional modifications for large agricultural environments.

8.4 Future Work

Several improvements can be implemented in future versions of the system.

8.4.1 Camera-Based Object Identification

A camera module and artificial intelligence algorithms can be integrated to identify:

- Humans
- Dogs
- Cats
- Birds
- Insects

This would improve threat classification accuracy.

8.4.2 IoT-Based Monitoring

Future systems can use:

- ESP32
- Wi-Fi Communication
- Cloud Platforms

for remote monitoring and data storage.

8.4.3 Mobile Application Integration

A mobile application can provide:

- Live monitoring
- Notification management
- Remote control
- Historical event records

8.4.4 Solar-Powered Operation

Solar panels can be integrated to support:

- Sustainable operation
- Extended deployment
- Reduced battery dependency

8.4.5 GPS-Based Navigation

Future versions may include GPS modules to enable:

- Outdoor navigation
- Location tracking
- Larger area monitoring

8.4.6 Cloud Data Logging

Cloud services can store:

- Motion detection records
- Alert history
- Sensor readings
- Performance statistics

This would support long-term agricultural analysis.

8.4.7 Advanced Obstacle Avoidance

Additional sensors such as:

- LiDAR
- Infrared Arrays
- Vision Systems

can improve navigation accuracy.

8.5 Final Conclusion

The introduction of the Smart Plant Monitoring Bot successfully presents an intelligent and autonomous plant monitoring and protection solution. It synthesizes line followed, turns, avoids obstacles, detects motion, communicates with remote location by GSM, displays information on the LCD screen, and emits an alarm; it is a total solution for smart farming application.

The results of the experiments obtained have proved that the robot has been able to patrol along a desired corridor, avoid obstacles, sense some activity around the plants, warn the user, and send SMS alerts. The system automatically minimizes manual monitoring and enhances plant protection efficiency.

While the prototype is a prototype for now, the project sets up a platform that offers potential potential for further developments under artificial intelligence, IoT connection, cloud computing, and in advanced navigation with the help of robots. As such, the Smart Plant Monitoring Bot could be an essential and helpful tool for automated plant monitoring and protection in modern farming conditions.

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Appendix A

Appendices

A Source Code / GitHub Link

The complete source code, project report, circuit diagrams, flowcharts, and other relevant resources of the **Smart Plant Monitoring Bot** are available through a GitHub repository for documentation, future development, and replication purposes.

GitHub Repository Link:

<https://github.com/Alif44-MT/Smart-Plant-Security-Bot-CSE-438-D2.git>

(Replace the above link with your actual GitHub repository URL.)

The main Arduino source code file is:

- `Smart_Plant_Monitoring_Bot.ino`

The source code includes the implementation of the following functionalities:

- IR sensor-based line following
- Ultrasonic sensor-based obstacle detection
- Servo motor-based path scanning
- Obstacle avoidance algorithm

- PIR motion detection for plant protection
- Buzzer alert system
- 16×2 I2C LCD status display
- SIM800L GSM SMS notification
- DC motor control using the L298N motor driver
- Automatic patrol recovery after obstacle or threat removal

For security reasons, sensitive information such as the mobile phone number, GSM configuration parameters, and any other private credentials should not be published directly in the public repository. These values should be replaced with placeholders before submission.

Example:

```
String phoneNumber = "YOUR_PHONE_NUMBER";
```

If future versions of the project include Wi-Fi or cloud-based monitoring (e.g., using ESP32 or ESP8266), sensitive information such as Wi-Fi SSID, Wi-Fi password, API keys, and cloud credentials should also be replaced with placeholders before making the repository public.

Example:

```
String ssid = "YOUR_WIFI_NAME";
String password = "YOUR_WIFI_PASSWORD";
String apiKey = "YOUR_API_KEY";
```

Table A.1: Team Member Responsibilities

Team Member	Contribution
Nishat Anjum	Arduino coding, line-follow logic, obstacle avoidance algorithm, GSM SMS integration, report writing, and software debugging.
Alif Mahmud Talha	Hardware setup, component selection, circuit connection, sensor calibration, system testing, system design, report writing, survey preparation, system analysis, requirement analysis, and PIR/buzzer integration support.

B Team Responsibilities

C. Ethics and Compliance Statements

The project was developed for academic and prototype demonstration purposes. The survey data was used only for academic research, and personal information was not shared in the report. Motion detection and distance data were used only for system testing.

The following ethical issues were considered:

- Testing results were reported honestly.
- Related works and technical documents were cited properly.
- Phone numbers and SMS content used for testing were kept private.
- The bot was tested safely in a controlled lab/garden environment.
- PIR and ultrasonic sensor data were used only for project demonstration.

In a real deployment, proper data privacy and user consent should be maintained.

D. Survey 1: Initial User Need Analysis

The first survey was conducted to understand user perception about the product concept and its possible usefulness. During the initial concept discussion, multiple plant protection approaches were considered. However, the final implemented project focused only on the Smart Plant Monitoring Bot.

Survey Questions

- What is your job role?
- Are you involved in purchasing or decision making?
- Have you understood the product concept?
- Do you think the product will be helpful for your workplace?
- Do you have any additional comments?

Table A.2: Summary of Survey 1 Response

Question / Area	Response Summary
Respondent Role	Doctor & Housewife
Decision Making Involvement	Yes
Understanding of Concept	Rated 4 out of 5.
Workplace Helpfulness	Rated 5 out of 5.
Additional Comment	The system is very helpful for gardening, but it will be more helpful if some features were added like: Camera, Control system by phone, add a water pump.

Interpretation of Survey 1

The first survey showed that the plant monitoring idea has practical value, but the concept needed clearer explanation regarding sensor reliability. The feedback helped the team focus on system stability, accurate obstacle detection, and proper hardware-software integration for the Smart Plant Monitoring Bot.

E. Survey 2: Prototype Feedback

The second survey was conducted after presenting the Smart Plant Monitoring Bot concept to a respondent from a gardening or agriculture-related background. The purpose was to evaluate product understanding, usefulness, purchase interest, price acceptability, and possible improvement areas.

Survey Questions

- What is your job role?
- Have you understood the product concept?
- Do you think the product will be helpful for your workplace?
- Would you purchase this product?
- Do you think the product price is acceptable?
- How much money would you spend for this type of product?
- Which part of this product needs improvement?
- Do you have any additional comments?

Table A.3: Summary of Survey 2 Response

Question / Area	Response Summary
Respondent Role	Teacher
Understanding of Concept	Rated 5 out of 5.
Workplace Helpfulness	Rated 4 out of 5.
Purchase Interest	Rated 4 out of 5.
Price Acceptability	Rated 4 out of 5.
Expected Budget	2000–2500 BDT
Improvement Suggestion	Add rechargeable option.
Additional Comment	Add some more features like add water pump, add rechargeable battery system and try to lower the price a little bit.

Interpretation of Survey 2

The second survey showed feedback about the Smart Plant Monitoring Bot's practical usefulness in home or garden settings. The respondent's understanding of the concept and purchase interest indicate the system's potential value. Improvement suggestions can guide future versions of the prototype.

F. Evidences of CEP

- **P1: Depth of Knowledge** — Required knowledge of embedded systems, robotics, IR sensors, ultrasonic ranging, servo control, and GSM communication.
- **P2: Conflicting Requirements** — Balanced low cost, reliable line detection, smooth obstacle avoidance, and real-time SMS alerting.
- **P3: Depth of Analysis** — Included analysis of IR sensor response, ultrasonic distance calculation, PIR motion behavior, and servo scan logic.
- **P4: Familiarity of Issues** — Line following is a known problem, but combining obstacle avoidance with PIR-based intruder alert created project-specific challenges.
- **P5: Applicable Standards** — Followed common Arduino hardware interfacing practices and used standard libraries.
- **P6: Stakeholder Needs** — Considered home plant owners, gardeners, and small farm caretakers.
- **P7: Interdependence** — IR sensors, ultrasonic, servo, PIR, buzzer, LCD, and GSM module were interconnected.

G. Evidences of CEA

- **A1: Range of Resources** — Used Arduino Uno, IR sensors, ultrasonic sensor, PIR sensor, servo motor, GSM module, LCD, L298N driver, Arduino IDE, and supporting libraries.

- **A2: Level of Interaction** — Multiple hardware and software modules interacted to complete line following, obstacle avoidance, and intruder alert operations.